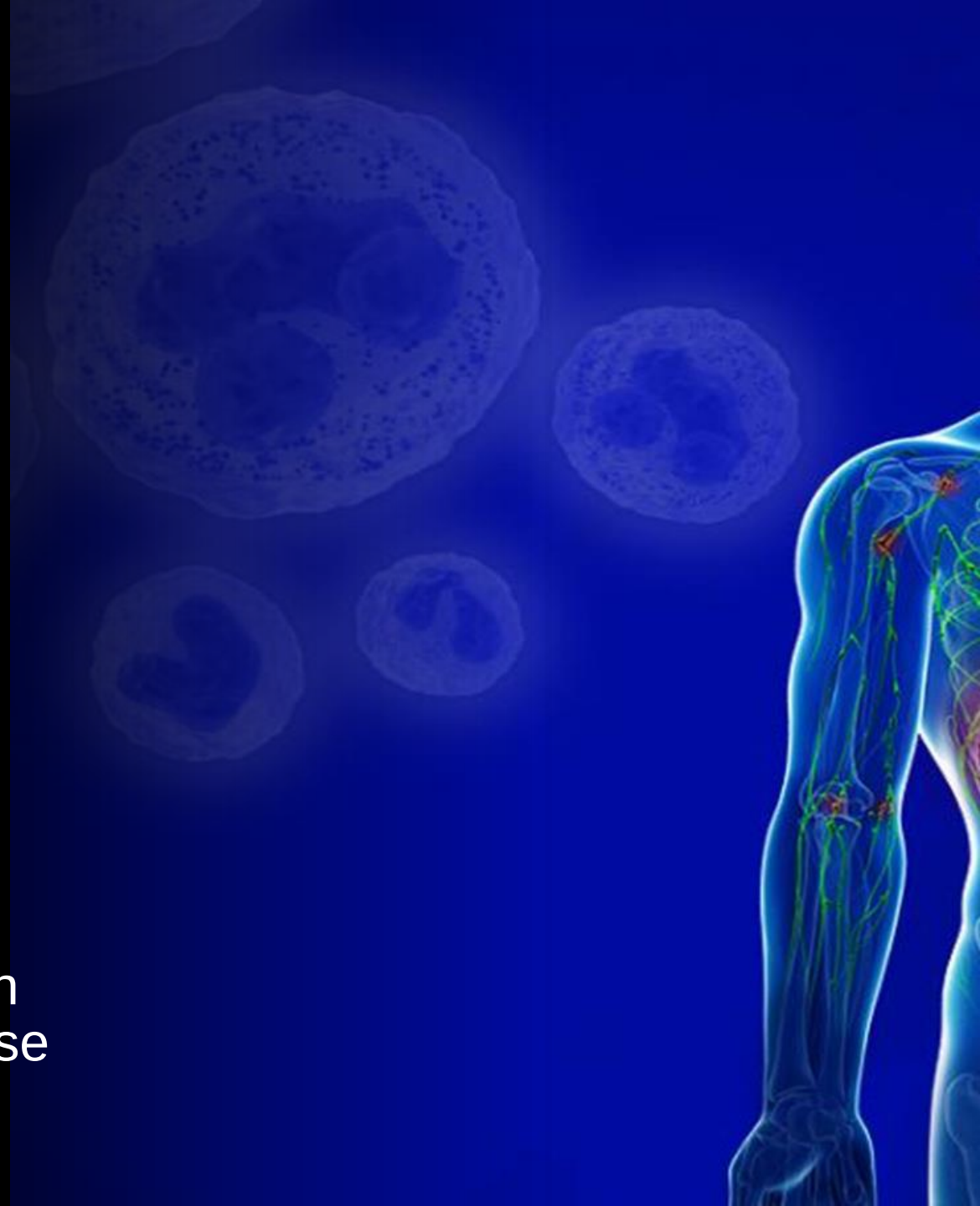
# PT7003: Inflammatory Joint Diseases

Irvind Sihota

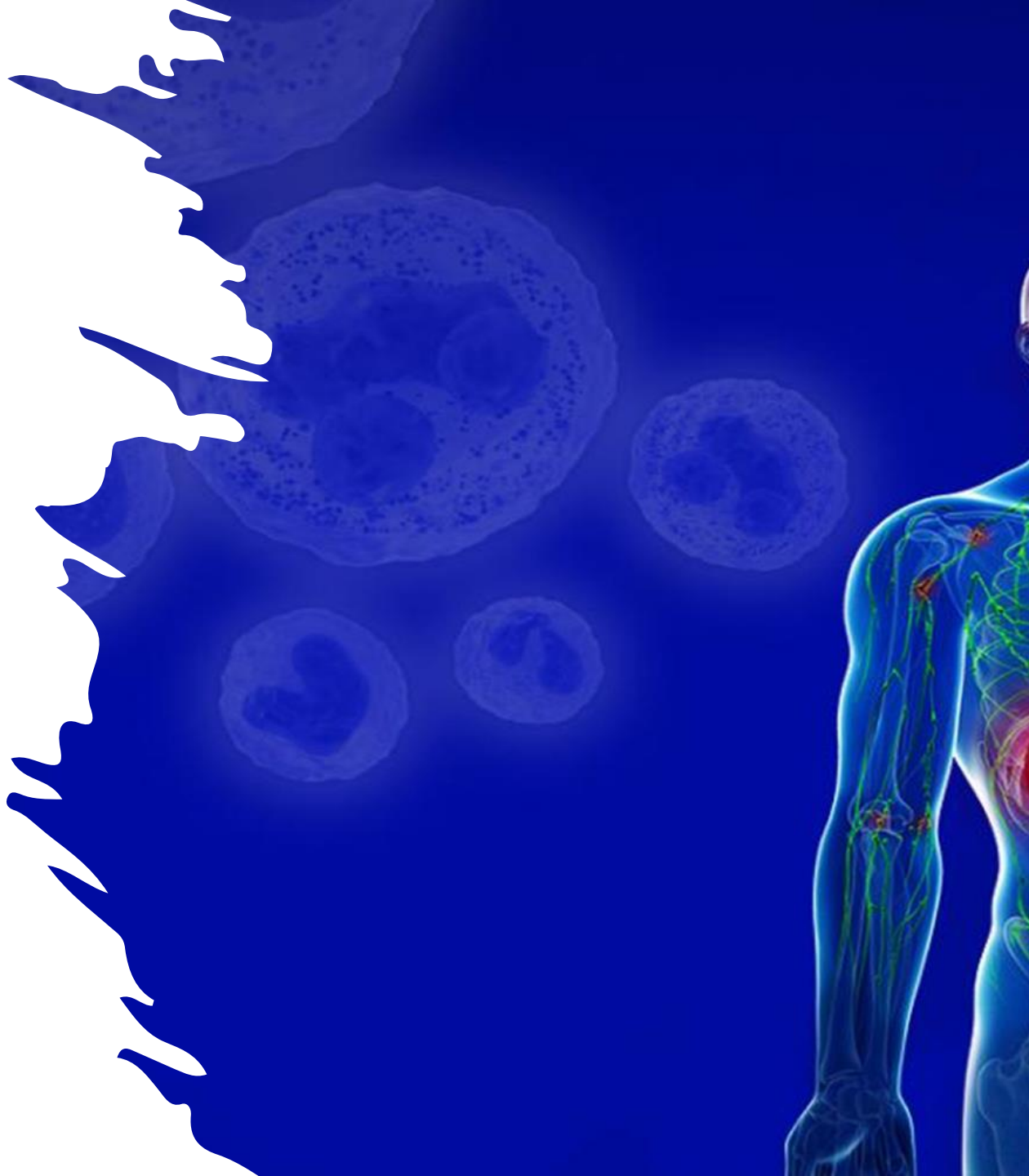
# LEARNING OUTCOMES

- Describe common syndromes that present with joint inflammation
- List the signs of symptoms of inflammatory joint disease and how these differ from degenerative conditions
- Recognise principal aspects of medical management of inflammatory joint disease
- Explain the ways in which physiotherapy can support people with inflammatory joint disease



# Inflammatory Joint Diseases

- Rheumatoid arthritis (RA),
- Axial Spondyloarthritis
  - Ankylosing spondylitis (AS)
- Psoriatic arthritis (PsA),
- Fibromyalgia
- Polymyalgia
- Systemic lupus erythematosus (SLE, lupus)
- Sjogrens
- Gout



# Why is a knowledge of inflammatory diseases important?

Your thoughts:



# Rheumatoid arthritis (RA)

Rheumatology originates from the Greek word ‘rheuma’ meaning that which ‘flows as a river or stream’.

Wilkinson, Ian B.; Raine, Tim; Wiles, Kate; Goodhart, Anna; Hall, Catriona; O'Neill, Harriet. Oxford Handbook of Clinical Medicine (Oxford Medical Handbooks) (p. 538). OUP Oxford. Kindle Edition.



# What is rheumatoid arthritis (RA)?

Rheumatoid arthritis, or RA, is an autoimmune and inflammatory disease, which means that your immune system attacks healthy cells in your body by mistake, causing inflammation (painful swelling) in the affected parts of the body.

RA mainly attacks the joints, usually many joints at once. RA commonly affects joints in the hands, wrists, and knees. In a joint with RA, the lining of the joint becomes inflamed, causing damage to joint tissue. This tissue damage can cause long-lasting or chronic pain, unsteadiness (lack of balance), and deformity (misshapenness).

RA can also affect other tissues throughout the body and cause problems in organs such as the lungs, heart, and eyes.

https://www.cdc.gov/arthritis/basics/rheumatoid-arthritis.html

# What is rheumatoid arthritis (RA)?

[redacted] autoimmune [redacted] immune system attacks healthy cells [redacted] by mistake, causing inflammation [redacted] in [redacted] parts of the body.

[redacted] mainly [redacted] joints, usually many joints [redacted]. [redacted] the hands, wrists, and knees. [redacted] lining [redacted] joint [redacted] inflamed, causing [redacted] tissue damage [redacted] long-lasting or chronic pain, unsteadiness (lack of balance), and deformity (misshapenness).

RA [redacted] tissues throughout [redacted] organs such as the lungs, heart, and eyes.

https://www.cdc.gov/arthritis/basics/rheumatoid-arthritis.html

# What is rheumatoid arthritis (RA)?

With RA, there are times when symptoms get worse, known as flares, and times when symptoms get better, known as remission.

Signs and symptoms of RA include:

Pain or aching in more than one joint
Stiffness in more than one joint
Tenderness and swelling in more than one joint
The same symptoms on both sides of the body (such as in both hands or both knees)
Weight loss
Fever
Fatigue or tiredness
Weakness

https://www.cdc.gov/arthritis/basics/rheumatoid-arthritis.html

# What is rheumatoid arthritis (RA)?

[redacted] flares, and [redacted] remission.

Signs and symptoms [redacted]:

Pain or aching in more than one joint
Stiffness in more than one joint
Tenderness and swelling in more than one joint
The same symptoms on both sides of the body (such as in both hands or both knees)
Weight loss
Fever
Fatigue or tiredness
Weakness

https://www.cdc.gov/arthritis/basics/rheumatoid-arthritis.html

# What is rheumatoid arthritis (RA)?

Characteristics that increase risk

- Age: Likelihood increases with age, highest onset is in the 60s
- Sex: New cases of RA are typically 2-3 times higher in women than men.
- Genetics/inherited traits: Inherited genes, called HLA (human leukocyte antigen) class II genotypes, can also make your arthritis worse. The risk of RA may be highest when people with these genes are exposed to environmental factors like smoking or when a person is obese.
- Smoking: increases a person's risk of developing RA and can make the disease worse.
- Early Life Exposures: Some early life exposures ***may*** increase risk of developing RA in adulthood. E.g., a study found that children whose mothers smoked had double the risk of developing RA as adults. Children of lower income parents are at increased risk of developing RA as adults.
- Obesity: Being obese can increase the risk of developing RA. Studies examining the role of obesity also found that the more overweight a person was, the higher his or her risk of developing RA became.

https://www.cdc.gov/arthritis/basics/rheumatoid-arthritis.html

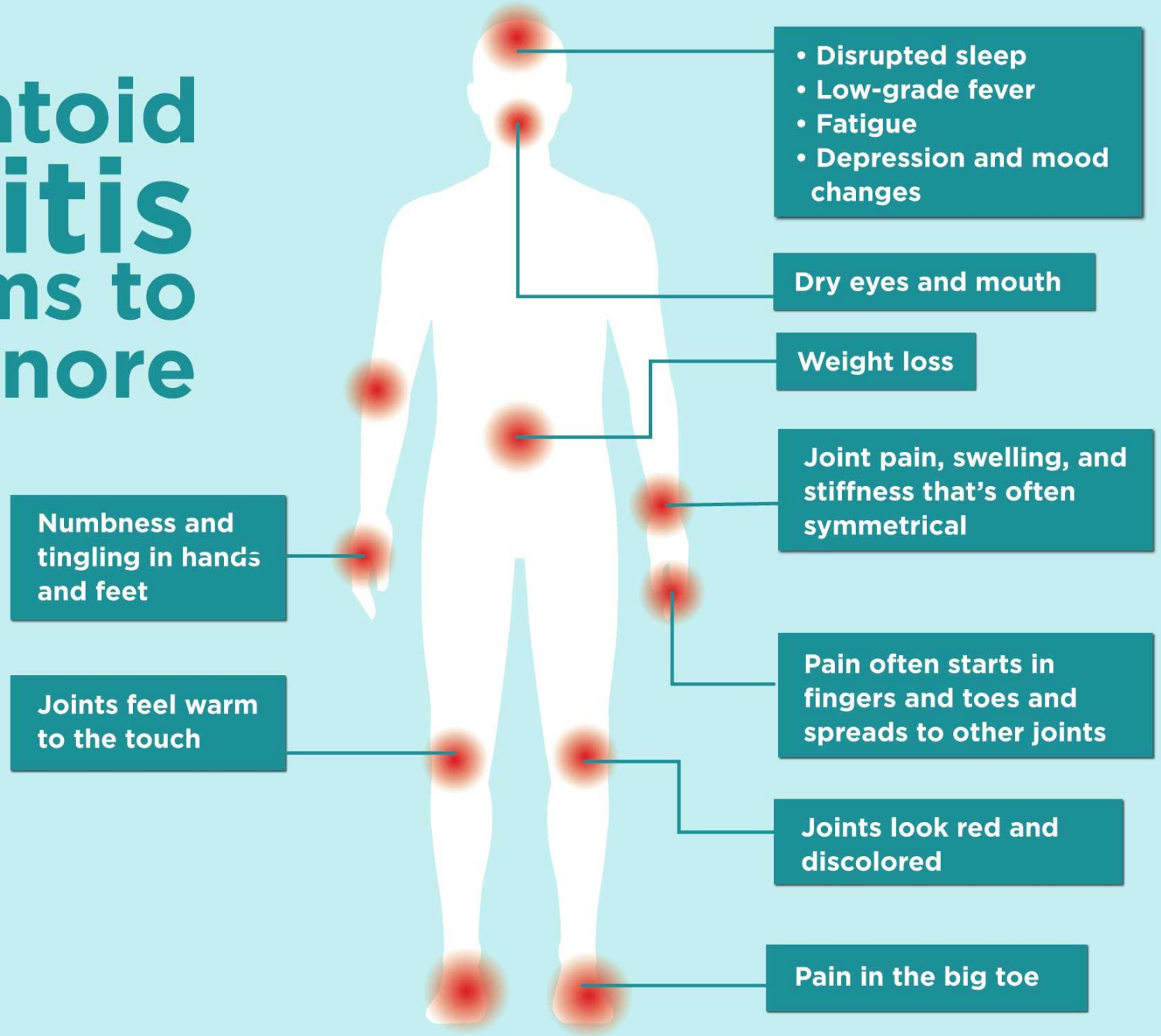
# What is rheumatoid arthritis (RA)?

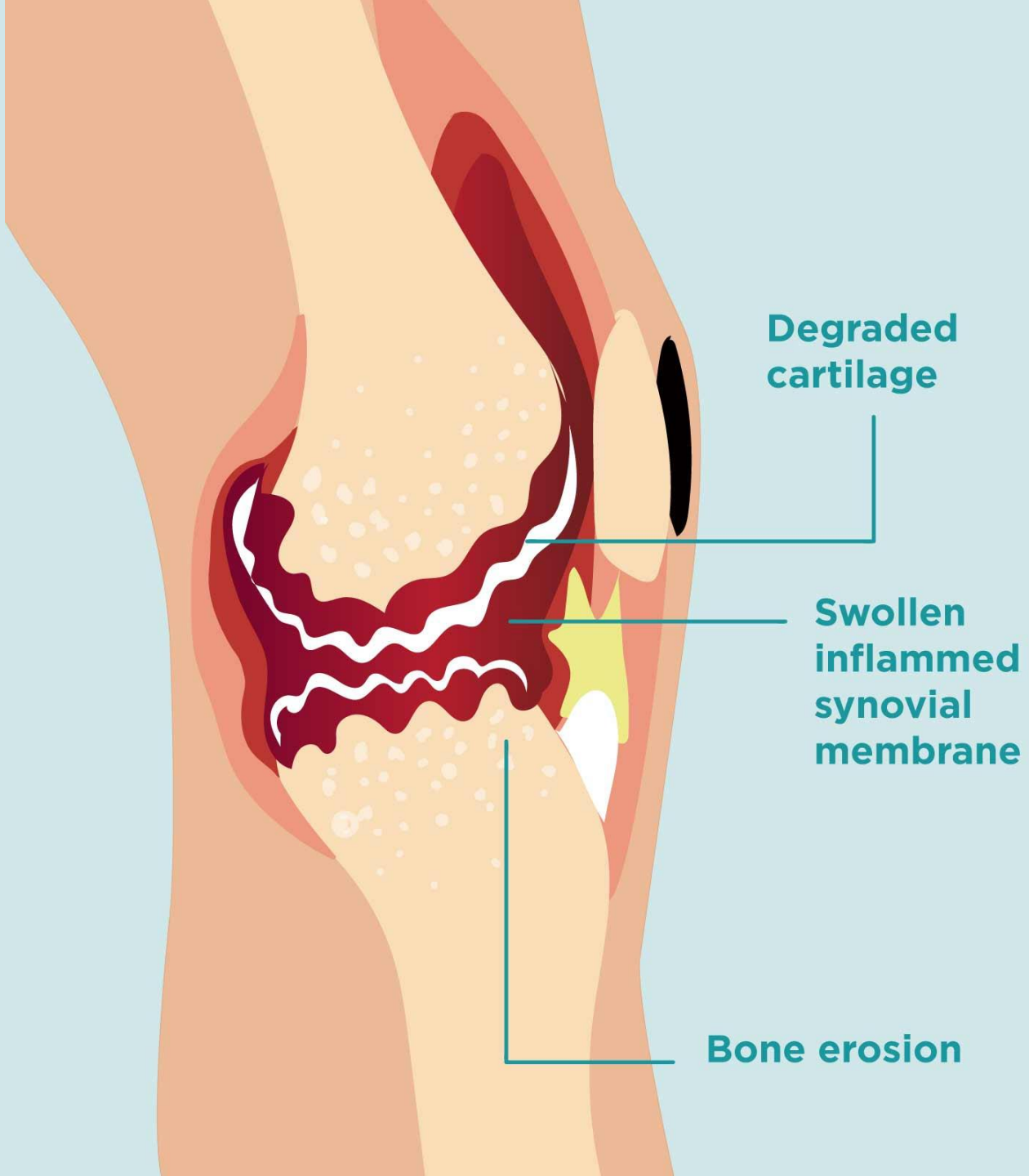
Characteristics that increase risk

- Age: Likelihood increases with age, highest onset is in the 60s
- Sex: [redacted] 2-3 times higher [redacted] women [redacted].
- [redacted] Inherited genes, called HLA (human leukocyte antigen) [redacted] can [redacted] make [redacted] arthritis worse. [redacted] risk [redacted] highest [redacted] exposed [redacted] environmental factors like smoking or [redacted] obese.
- [redacted]
- [redacted] early life exposures ***may*** increase risk of [redacted] RA in adulthood. [redacted] children whose mothers smoked [redacted] double [redacted] RA as adults. Children [redacted] lower income parents [redacted] risk of developing RA as adults.
- Obesity [redacted] can increase [redacted] risk of [redacted] RA. [redacted]

https://www.cdc.gov/arthritis/basics/rheumatoid-arthritis.html

# Rheumatoid Arthritis symptoms to never ignore






# The Stages of Rheumatoid Arthritis

### STAGE 1

The body mistakenly attacks its own joint tissue.

### STAGE 2

The body makes the antibodies and the joints start swelling up.

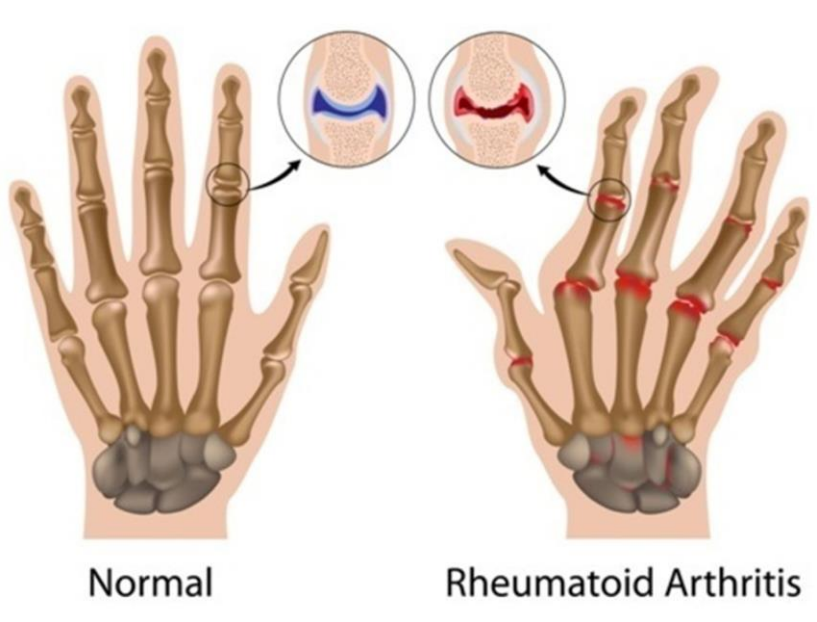
### STAGE 3

The joints start becoming bent and deformed, the fingers become crooked. These misshapen joints can press on the nerves and can cause nerve pain as well.

### STAGE 4

If not treated, the disease will progress to the last stage, in which there's no joint remaining at all and the joint is essentially fused.

# What is rheumatoid arthritis (RA)?










# RA Management

Management

- Refer early to a rheumatologist (before irreversible destruction). They will consider use of medications.
- Physiotherapy and occupational therapy, eg for aids and splints.
- Surgery may relieve pain, improve function, and prevent deformity.

Wilkinson, Ian B.; Raine, Tim; Wiles, Kate; Goodhart, Anna; Hall, Catriona; O'Neill, Harriet. Oxford Handbook of Clinical Medicine (Oxford Medical Handbooks) (p. 546). OUP Oxford. Kindle Edition.

# Axial Spondyloarthritis
# Ankylosing spondylitis (AS)

Axial (from Latin axōn, meaning "axle"): around the central axis of the organism or the extremity.
https://operativeneurosurgery.com/doku.php?id=axial#:~:text=Axial%20(from%20Latin%20ax%C5%8Dn%2C%20meaning,axis%20of%20an%20organism%2C%20respectively.

spondylitis (n.)
"inflammation of the vertebrae," 1837, Modern Latin, from Latin spondylus, from Greek spondylos (see spondylo-) + -itis "inflammation."
https://www.etymonline.com/word/spondylo-

ankylosis (n.)
"stiffening of joints caused by consolidation or fusion of two or more bones into one," 1713, alternative (and more etymological) spelling of anchylosis (q.v.).
https://www.etymonline.com/search?q=ankylo



# Axial Spondyloarthritis

**axSpA, axial SpA, radiographic axial Spondyloarthritis** (r-axSpA)

https://youtu.be/oNi5e5RCfrs

So, although the pathophysiology of axSpA remains incompletely understood, the progress in recent years in several fields of research in axSpA including genetics, diagnosis, imaging and therapeutics, hold great promise for the future.

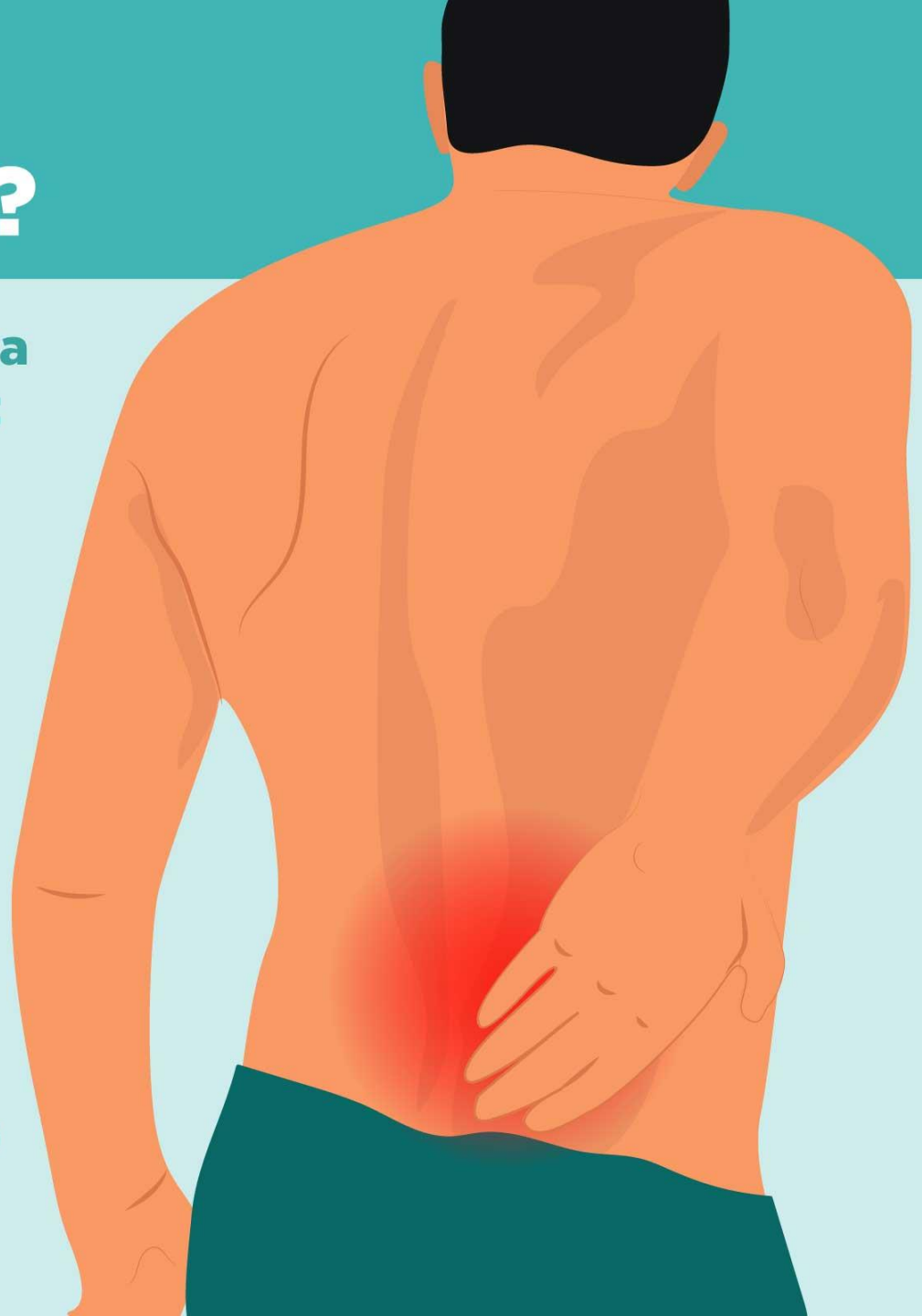
# What Is Axial Spondyloarthritis?

**Axial spondyloarthritis (axSpA) is a type of inflammatory arthritis that causes lower back pain, among other symptoms.**

**There are two main types:**

- **Non-radiographic axial spondyloarthritis**
- **Ankylosing spondylitis**

**AxSpA can be debilitating and progressive, so seeing a rheumatologist for early diagnosis and treatment is key.**



# Axial Spondyloarthritis

1. Seronegativity (= rheumatoid factor −ve).
2. HLA B27 association—see BOX.
3. 'Axial arthritis': pathology in spine (spondylo-) and sacroiliac joints.
4. Asymmetrical large-joint oligoarthritis (ie <5 joints) or monoarthritis.
5. Enthesitis: inflammation of the site of insertion of tendon or ligament into bone, eg plantar fasciitis, Achilles tendonitis, costochondritis.
6. Dactylitis: inflammation of an entire digit ('sausage digit'), due to soft tissue oedema, and tenosynovial and joint inflammation.
7. Extra-articular manifestations: eg iritis (anterior uveitis), psoriaform rashes (psoriatic arthritis), oral ulcers, aortic valve incompetence, inflammatory bowel disease.

# Ankylosing Spondylitis

Ankylosing spondylitis (AS) A chronic inflammatory disease of the spine and sacroiliac joints, of unknown aetiology
Prevalence: 0.25–1%. Men present earlier: ~6:1 (male:female) at 16yrs old, and 2:1 (male:female) at 30yrs old. 90% are HLA B27 +ve
Symptoms and signs: The typical patient is a man <30yrs old - gradual onset of low back pain, worse during the night with spinal morning stiff ness relieved by exercise. Pain radiates from sacroiliac joints to hips/buttocks, and usually improves towards the end of the day.
Progressive loss of spinal movement (all directions)— including thoracic expansion.
Disease course is variable; a few progress to kyphosis, neck hyperextension (question-mark posture; fig 12.13), and spino-cranial ankylosis.
Enthesitis - especially Achilles tendonitis, plantar fasciitis, at the tibial and ischial tuberosities, and at the iliac crests.
Anterior mechanical chest pain due to costochondritis and fatigue may feature.

# Ankylosing Spondylitis

[redacted]chronic inflammatory disease [redacted]spine and sacroiliac joints[redacted]unknown aetiology
Prevalence: 0.25–1%. Men[redacted]earlier: ~6:1 (male:female) [redacted]16yrs [redacted], [redacted]2:1 (male:female) at 30yrs [redacted]90% are HLA B27 +ve
[redacted]typical patient [redacted]man <30yrs old - gradual onset of low back pain, worse [redacted][redacted]night [redacted]spinal morning stiffness relieved by exercise. Pain radiates [redacted]sacroiliac joints to hips/buttocks, [redacted]improves [redacted]end of the day.
Progressive loss of spinal movement (all directions)— including thoracic expansion.
Disease [redacted]variable; [redacted]to kyphosis, neck hyperextension [redacted]and spino-cranial ankylosis.
Enthesitis - [redacted]Achilles tendonitis, plantar fasciitis, [redacted]tibial [redacted]ischial tuberosities, [redacted]iliac crests.
Anterior mechanical chest pain [redacted]costochondritis [redacted]fatigue [redacted].

# Ankylosing Spondylitis

- Acute iritis occurs in ~⅓ of patients and may lead to blindness if untreated (but may also have occurred many years before, so enquire directly).
- AS is also associated with osteoporosis (up to 60%), aortic valve incompetence (<3%), and pulmonary apical fibrosis.
- MRI allows detection of active inflammation (bone marrow oedema) as well as destructive changes such as erosions, sclerosis, and ankylosis.
- X-rays can show SI joint space narrowing or widening, sclerosis, erosions, and ankylosis/fusion. Vertebral syndesmophytes are characteristic (often T11–L1 initially): bony proliferations due to enthesitis between ligaments and vertebrae. These fuse with the vertebral body above, causing ankylosis.
- In later stages, calcification of ligaments with ankylosis lead to a 'bamboo spine' appearance.
- Wilkinson, Ian B.; Raine, Tim; Wiles, Kate; Goodhart, Anna; Hall, Catriona; O'Neill, Harriet. Oxford Handbook of Clinical Medicine (Oxford Medical Handbooks) (p. 550). OUP Oxford. Kindle Edition.

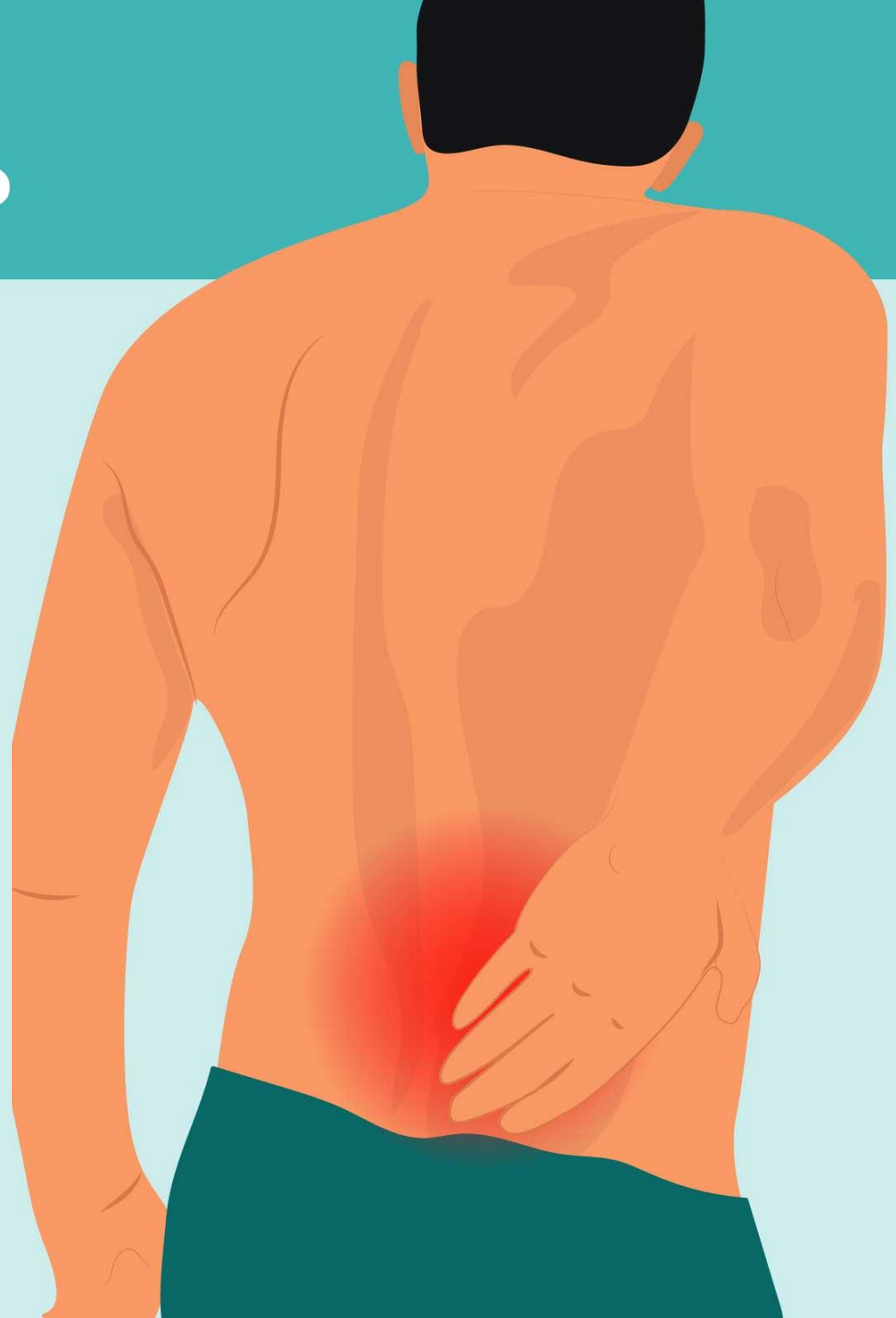
# Ankylosing Spondylitis

- [redacted] iritis [redacted] ~⅓ of patients [redacted] blindness if untreated [redacted]
- AS [redacted] osteoporosis (up to 60%), aortic valve incompetence (<3%), and pulmonary apical fibrosis.
- MRI [redacted] active inflammation (bone marrow oedema) [redacted] destructive changes [redacted] as erosions, sclerosis, and ankylosis.
- X-rays [redacted] SI joint space narrowing or widening, sclerosis, erosions, and ankylosis/fusion. [redacted] syndesmophytes [redacted] bony proliferations [redacted] enthesitis [redacted] ligaments [redacted] vertebrae. [redacted] fuse [redacted] vertebral [redacted] causing ankylosis.
- In later [redacted] calcification [redacted] ligaments with ankylosis [redacted] a ‘bamboo spine’
- [redacted]



# What Is Axial Spondyloarthritis?

Management: Exercise, not rest for back pain. Exercise programmes to maintain posture and mobility (including thoracic expansion).

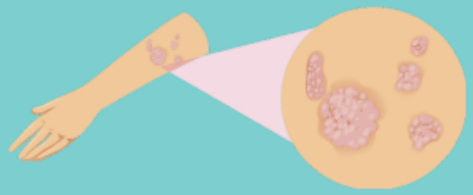


# SCREEND'EM
## BEFORE YOU TREAT'EM

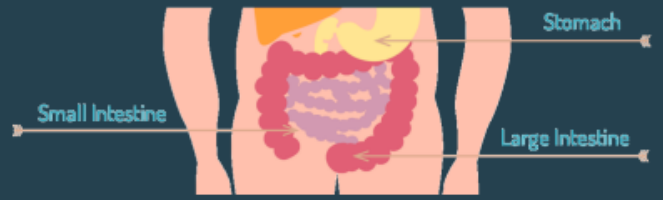
**A clinical tool to help identify spondyloarthropathy (SpA)**

## SKIN

**6-42% of patients with psoriasis develop psoriatic arthritis.**



## COLITIS OR CROHN'S




**Arthritis is one of the most common extra-intestinal manifestations of inflammatory bowel disease. The prevalence of SpA in patients with Crohn's is estimated to be 26% at 6 year follow up.**

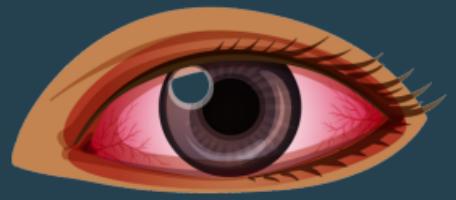
# RELATIVES



There is a strong relationship between SpA and HLA-B27 positive patients.

Family members of patients with SpA who are HLA-B27 positive have a 16-fold increase chance of developing ankylosing spondylitis if they are also HLA-B27 positive.

# EYES

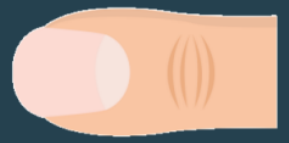


Acute anterior uveitis (AAU) can cause a painful, red eye with photophobia and blurred vision. 40% of patients presenting with idiopathic AAU have undiagnosed SpA. 50% of patients with AAU are HLA-B27 positive and >50% of these have SpA.

# EARLY MORNING STIFFNESS

Inactivity related stiffness that lasts for more than 30 minutes is suggestive of inflammatory disease.

# NAILS



Nail lesions occur in 87% of SpA patients and include:

- small depressions in the nail (pitting)
- thickening of the nails
-painless detachment from the nail bed (onchylosis).

# DACTYLITIS

Sausage like swelling of the digits is a hallmark sign of psoriatic arthritis, occuring in 50% of cases.



# ENTHESITIS

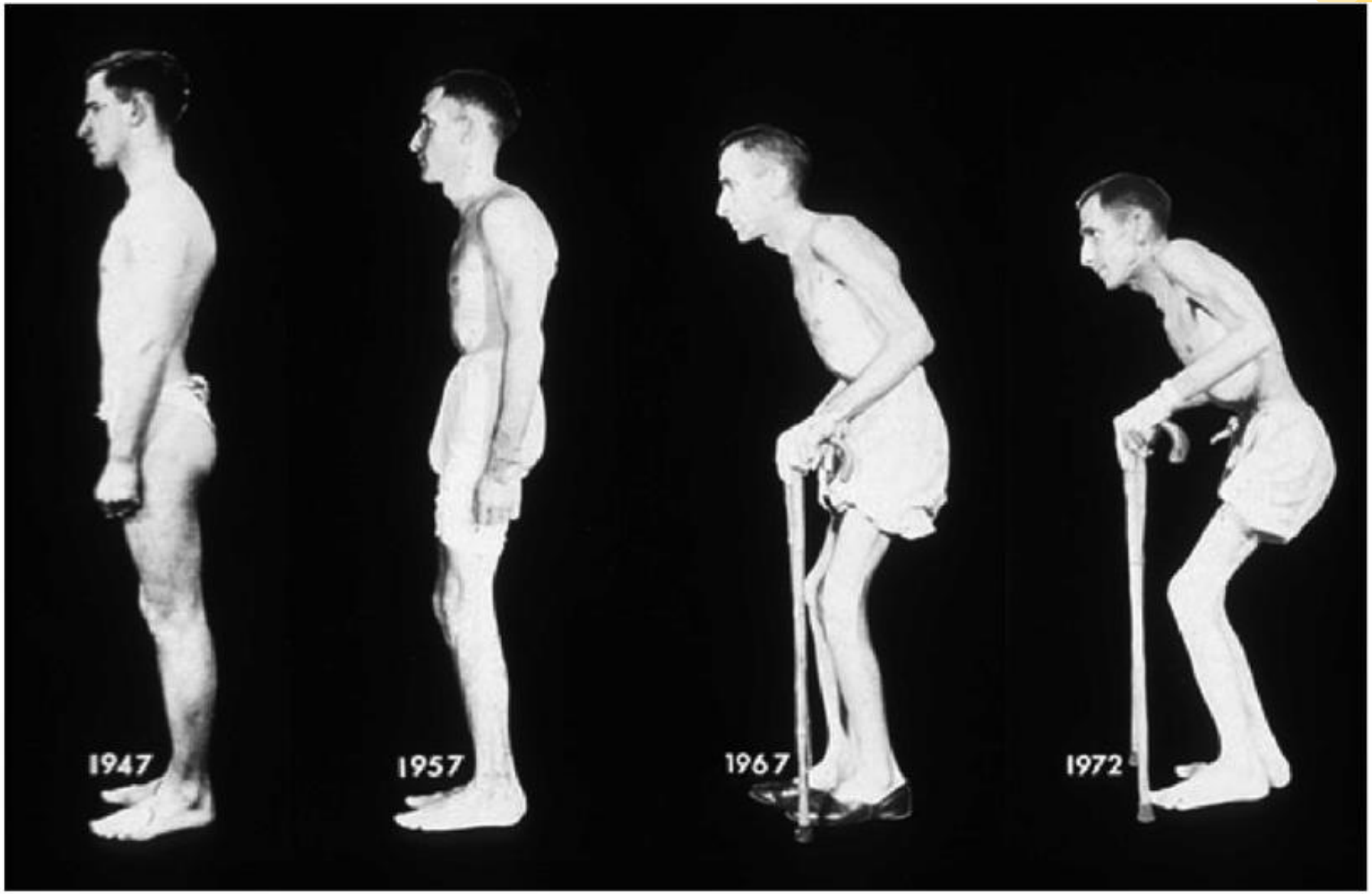
98% of SpA patients have at least one abnormal enthesis. The most common sites are the Achilles tendon, plantar fascia and patellar tendon.



# MOVEMENT & MEDICATION EFFECT

SpA patients report improvement with activity but not with rest, and a favourable response to NSAIDs.

# Ankylosing Spondylitis

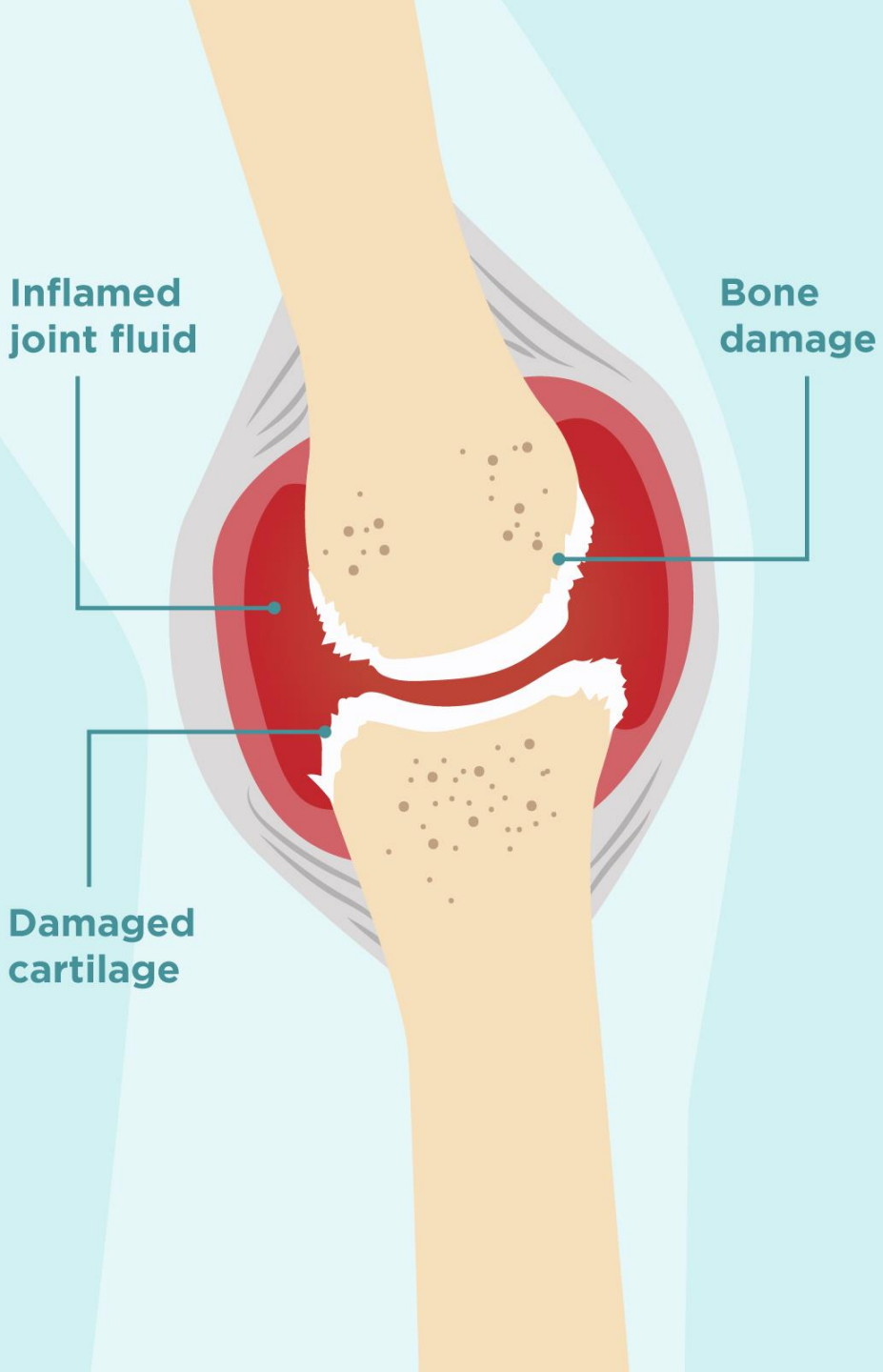


# Psoriatic arthritis (PsA)

Psoriatic arthritis (OHCS p594.)
Occurs in 10–40% with psoriasis and can present  before skin changes. 1:1 (male to female, 10 – 20, 30 – 50). HLA B27 – 85%
Patterns are:
•Early morning stiffness.
•symmetrical polyarthritis (like RA)
•DIP joints
•asymmetrical oligoarthritis
•spinal (similar to AS)
•Radiology: Erosive changes, with ‘pencil-in-cup’  deformity in severe cases. Associated with nail changes in 80%, synovitis (dactylitis), acneiform rashes and palmo-plantar pustulosis.



Wilkinson, Ian B.; Raine, Tim; Wiles, Kate; Goodhart, Anna; Hall, Catriona; O'Neill, Harriet. Oxford Handbook of Clinical Medicine (Oxford Medical Handbooks) (p. 551). OUP Oxford. Kindle Edition.




# Stages of Psoriatic Arthritis

## Pre-Clinical Psoriatic Arthritis

Inflammation in the body leads to a variety of early symptoms often in people who already have psoriasis, such as fatigue, heel pain, and joint pain.

## Early Psoriatic Arthritis

Different PsA “domains” can cause variation in symptoms from person to person. These include joint pain and stiffness, back pain, nail changes, psoriasis skin rash, finger and toe swelling, and tendon/ligament pain (enthesitis) at the Achilles tendon, bottom of the foot (plantar fasciitis), or elbow (tennis elbow).

## Long-term Psoriatic Arthritis

If inflammation is not under control, PsA can get worse in any of the different domains. Advanced PsA may cause frequent flares, loss of joint mobility, permanent bone damage, and damage to other organs.

GHLF creakyjoints®

# Fibromyalgia and chronic fatigue syndrome

Fibromyalgia and chronic fatigue syndrome are part of a diffuse group of overlapping syndromes, sharing similar demographic and clinical characteristics, in which chronic symptoms of fatigue and widespread pain feature prominently. Their existence as discrete entities is controversial, especially in the absence of clear pathology, and some find such dysfunctional diagnoses frustrating.

## Fibromyalgia

*Prevalence:* 0.5–4%. ♀:♂≈6:1 but varies depending on which diagnostic criteria are used .

*Risk factors:* Female sex, middle age, low household income, divorced, low educational status.

*Associations:* Other somatic syndromes such as chronic fatigue syndrome, irritable bowel syndrome (p266), and chronic headaches syndromes Also found in ~25% of patients with RA, AS, and SLE.

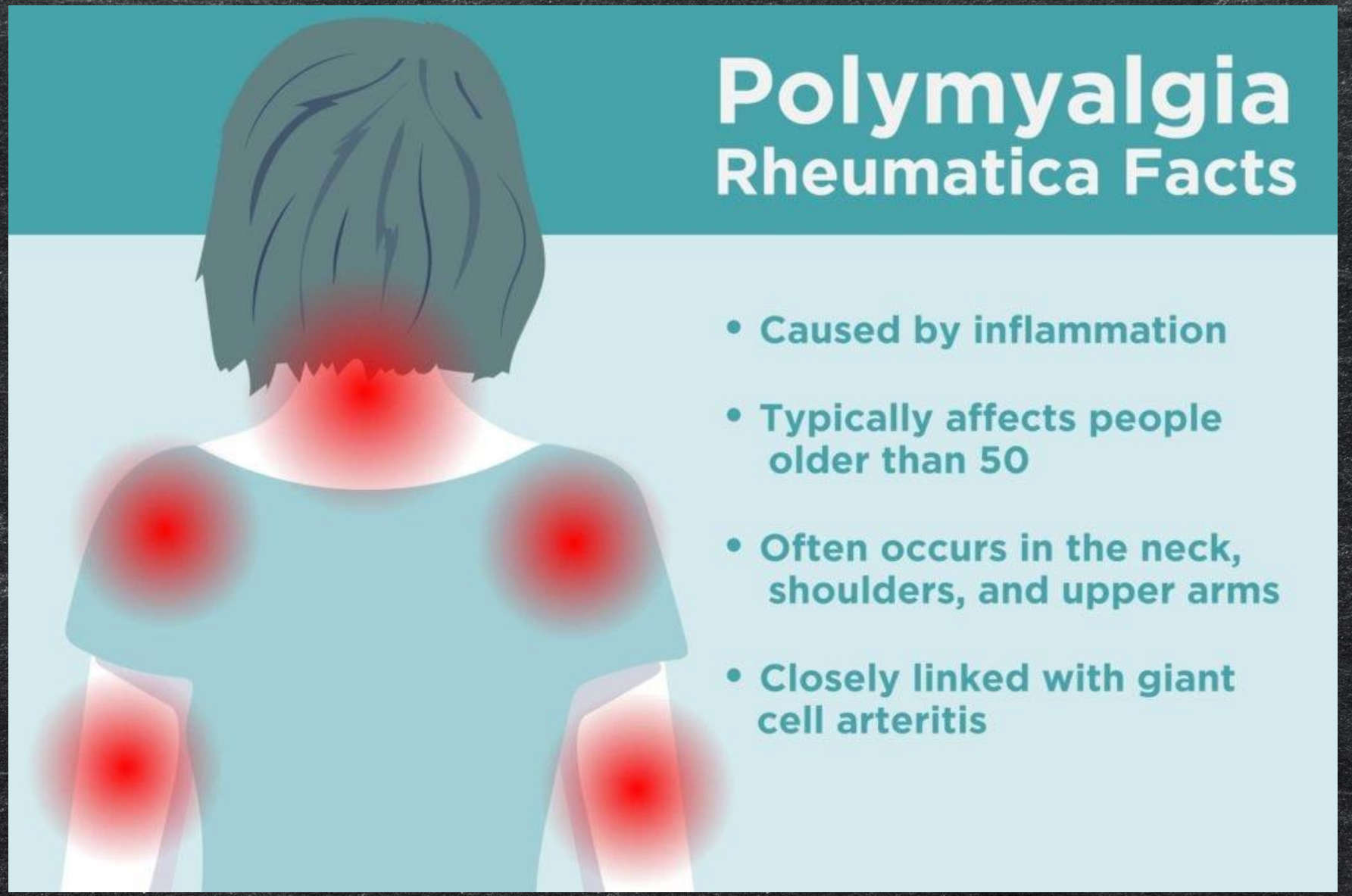
*Features:* Diagnosis depends on pain that is *chronic* (>3 months) and widespread (involves left and right sides, above and below the waist, and the axial skeleton). Profound fatigue is almost universal with complaint of unrefreshing sleep and significant fatigue and pain with small increases in physical exertion. *Additional features:* morning stiffness (~80–90%), paraesthesiae (without underlying cause), headaches (migraine and tension), poor concentration, low mood, and sleep disturbance (~70%). Widespread and severe tender points.

*Investigations:* All normal. Diagnosis is clinical. Over-investigation can consolidate illness behaviour; but exclude other causes of pain and/or fatigue (eg RA, PMR see p557, vasculitis see p556, hypothyroidism see p220, myeloma see p368).

# Fibromyalgia

*Management:* Multidisciplinary with optimal results coming from full engagement of the patient who should be encouraged to remain as active as they feel able, and ideally to continue to participate in the workforce. New symptoms should be fully reviewed to exclude an alternative diagnosis. Patients should be advised that there is no one specific treatment that is guaranteed to work, but any of the following may help: graded exercise programmes, including both aerobic and strength-based training. Pacing of activity is vital to avoid over-exertion and consequent pain and fatigue. Long-term *graded exercise programmes* improve functional capacity.[55,56] Relaxation, rehabilitation and physiotherapy may also help. *Cognitive-behavioural therapy* (CBT) aims to help patients develop coping strategies and set achievable goals.[57]

# Polymyalgia



# Polymyalgia

## Polymyalgia rheumatica (PMR)

*Features:* Age >50yrs; subacute onset (<2 weeks) of bilateral aching, tenderness, and morning stiffness in shoulders, hips, and proximal limb muscles ± fatigue, fever, ↓weight, anorexia, and depression. There may be associated mild polyarthritis, tenosynovitis, and carpal tunnel syndrome (10%). Weakness is not a feature.

# Systemic lupus erythematosus



What is SLE?

Systemic lupus erythematosus (SLE), is the most common type of lupus. SLE is an autoimmune disease in which the immune system attacks its own tissues, causing widespread inflammation and tissue damage in the affected organs (can affect the joints, skin, brain, lungs, kidneys, and blood vessels).

There is no cure for lupus, but medical interventions and lifestyle changes can help control it.

What causes SLE?: The causes of SLE are unknown, but are believed to be linked to environmental, genetic, and hormonal factors.

What are the signs and symptoms?

People with SLE may experience a variety of symptoms that include fatigue, skin rashes, fevers, and pain or swelling in the joints. Among some adults, having a period of SLE symptoms—called flares—may happen every so often, sometimes even years apart, and go away at other times—called remission. However, other adults may experience SLE flares more frequently throughout their life.

https://www.cdc.gov/lupus/facts/detailed.html

What is SLE?

What are the complications of SLE?

SLE can have both short- and long-term effects on a person's life.

Early diagnosis and effective treatments can help reduce damage, improve function and Quality of Life.

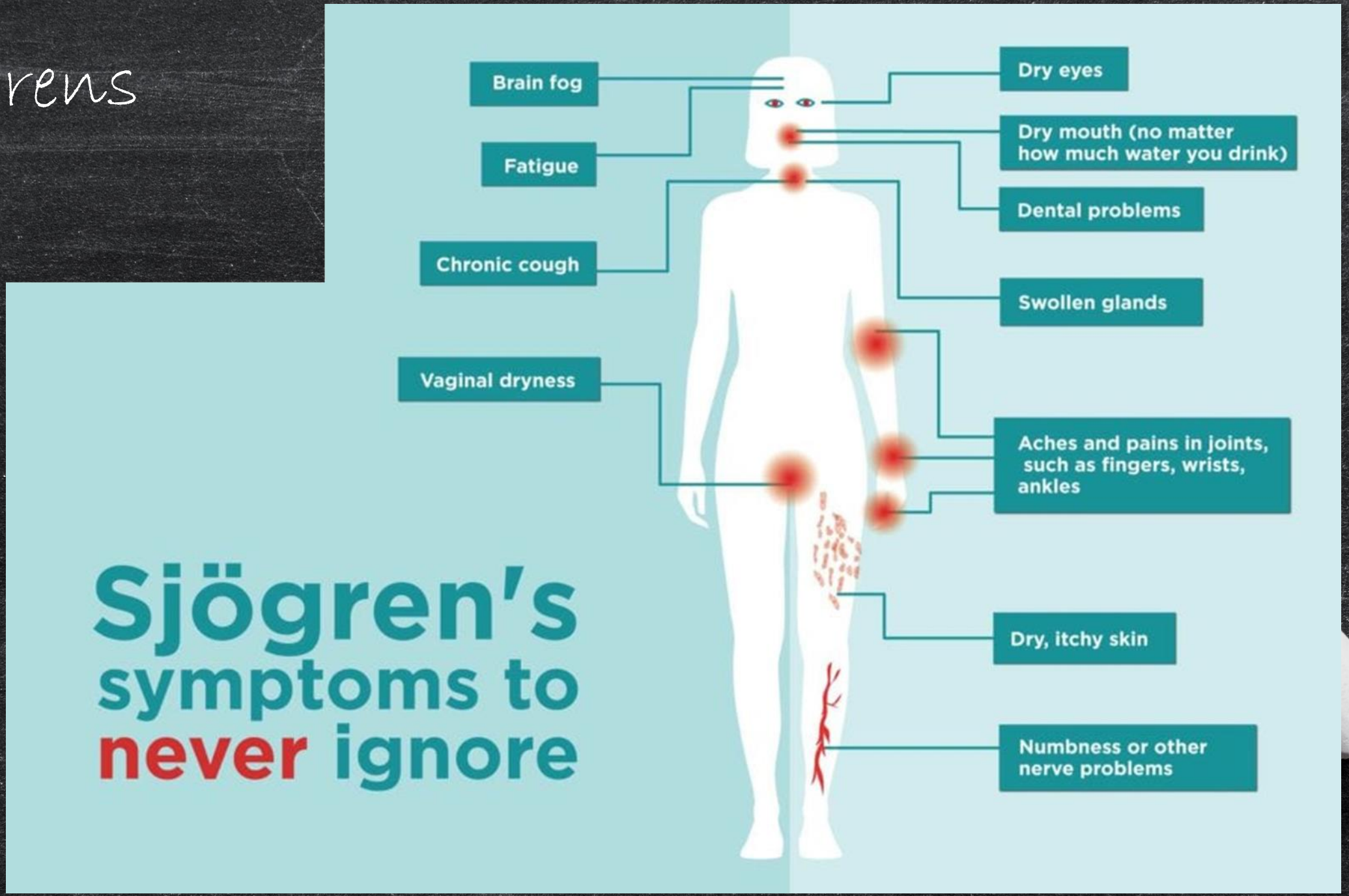
Poor care, late diagnosis, less effective treatments may increase the damaging effects of SLE, causing more complications and an increased risk of death.

SLE can limit a person's physical, mental, and social functioning. These limitations experienced by people with SLE can impact their quality of life, especially if they experience fatigue.

Fatigue is the most common symptom negatively affecting the quality of life of people with SLE.

https://www.cdc.gov/lupus/facts/detailed.html

# sjogrens


Brain fog
Fatigue
Chronic cough
Vaginal dryness
Dry eyes
Dry mouth (no matter how much water you drink)
Dental problems
Swollen glands
Aches and pains in joints, such as fingers, wrists, ankles
Dry, itchy skin
Numbness or other nerve problems
Sjögren's symptoms to never ignore

# Sjogren’s

Sjogren's (SHOW-grins) syndrome is a disorder of your immune system identified by its two most common symptoms — dry eyes and a dry mouth.

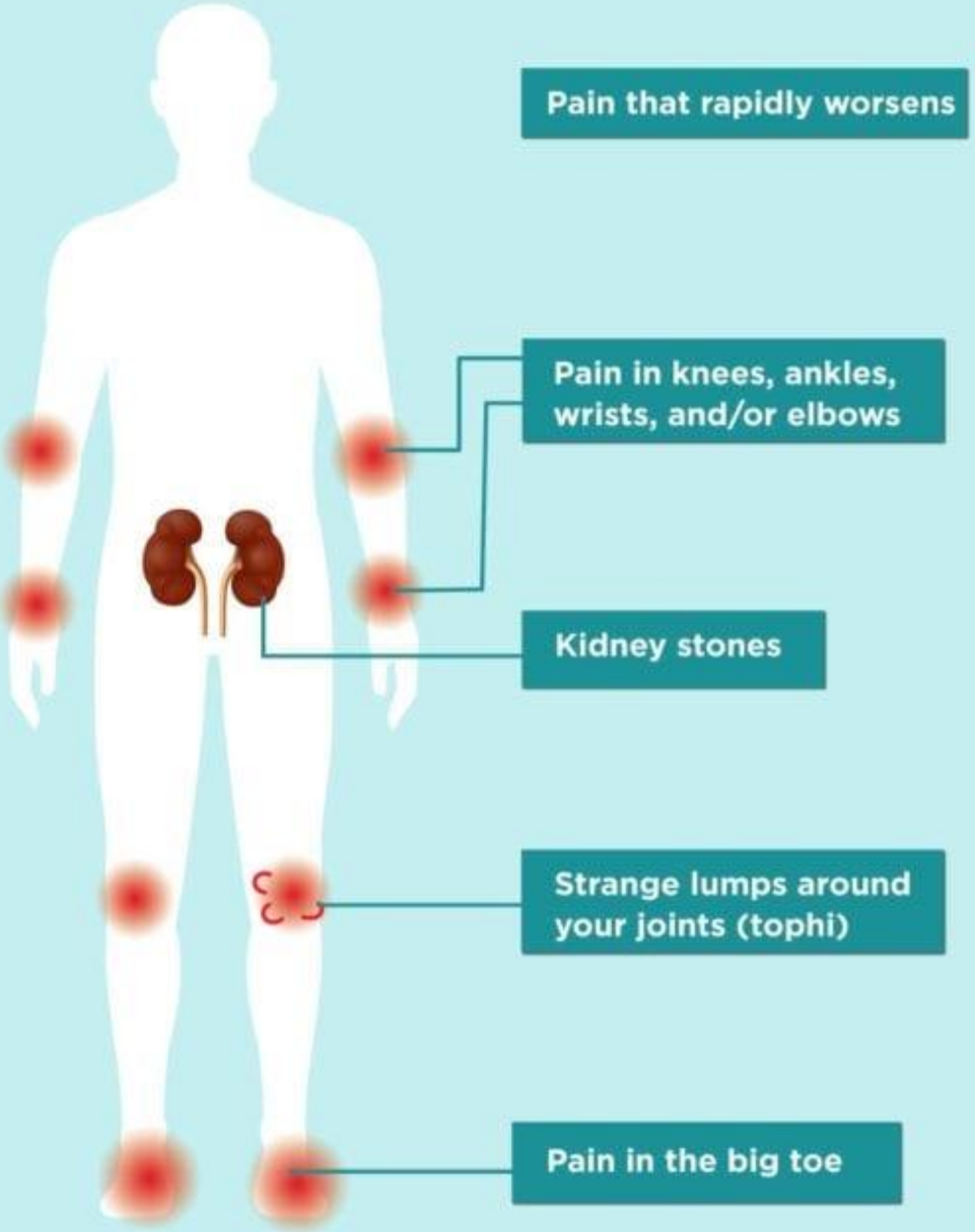
The condition often accompanies other immune system disorders, such as rheumatoid arthritis and lupus. In Sjogren's syndrome, the mucous membranes and moisture-secreting glands of your eyes and mouth are usually affected first — resulting in decreased tears and saliva.

Although you can develop Sjogren's syndrome at any age, most people are older than 40 at the time of diagnosis. The condition is much more common in women. Treatment focuses on relieving symptoms.

https://www.mayoclinic.org/diseases-conditions/sjogrens-syndrome/symptoms-causes/syc-20353216

Gout

# Gout
## symptoms to never ignore



# Gout

## Crystal arthropathies: gout

Gout[31] typically presents with an acute monoarthropathy with severe joint inflammation (fig 12.8). >50% occur at the metatarsophalangeal joint of the big toe (fig 12.12) (podagra). Other common joints are the ankle, foot, small joints of the hand, wrist, elbow, or knee. It can be polyarticular. It is caused by deposition of monosodium urate crystals in and near joints. Attacks may be precipitated by trauma, surgery, starvation, infection, or diuretics. It is associated with raised plasma urate. In the long term, urate deposits (= tophi, eg in pinna, tendons, joints; see fig 12.9) and renal disease (stones, interstitial nephritis) may occur. *Prevalence:* ~1%. ♂:♀ ≈ 4:1.

# Gout



Fig 12.8 Acute monoarthritis in gout.



Fig 12.9

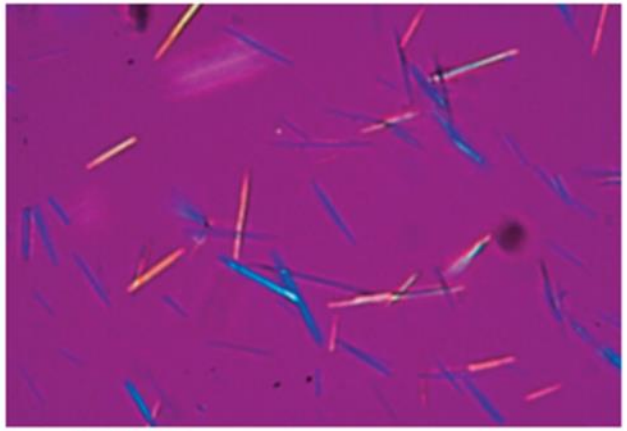


Fig 12.10 **N**eedle-shaped monosodium urate crystals found in gout, displaying **N**egative birefringence under polarized light.

Reproduced from Warrell *et al.*, *Oxford Textbook of Medicine*, 2010, with permission from Oxford University Press.

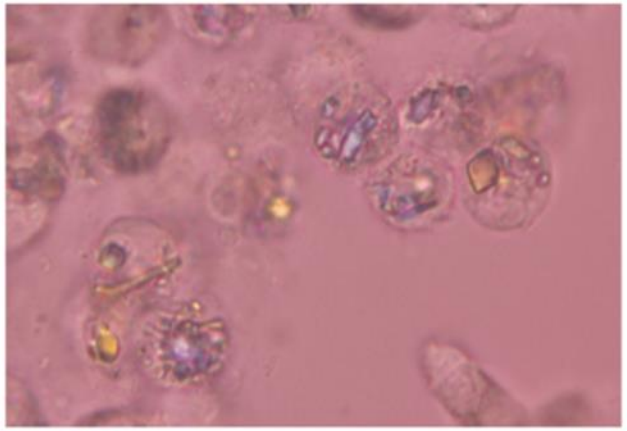
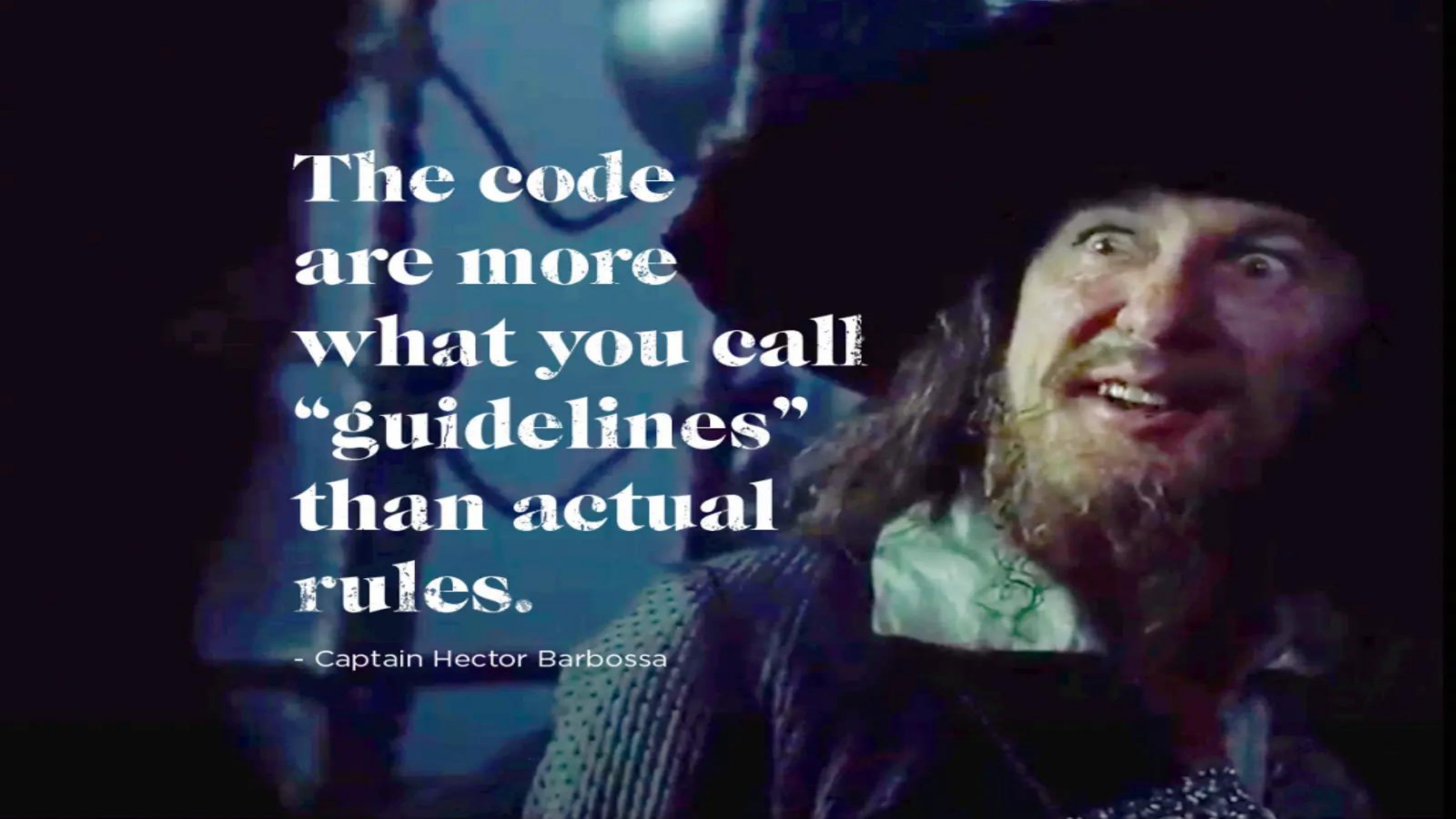


Fig 12.11 Rhomboid-shaped calcium pyrophosphate dihydrate crystals in **P**seudogout, showing **P**ositive birefringence in polarized light.

Image courtesy of Prof. Eliseo Pascual, Sección de Reumatología, Hospital General Universitario de Alicante.


The code are more what you call "guidelines" than actual rules.
- Captain Hector Barbossa

# Differentiation

## *Presenting symptoms:*

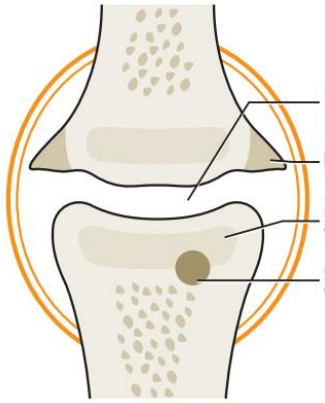
- Pattern of involved joints.
- Symmetry (or not).
- Morning stiffness >30min (eg RA).
- Pain, swelling, loss of function, erythema, warmth.

| Monoarthritis | Oligoarthritis (≤5 joints) | Polyarthritis (>5 joints involved) | |
|---|---|---|---|
| Septic arthritis | Crystal arthritis | **Symmetrical** | **Asymmetrical** |
| Crystal arthritis (gout, CPPD) | Psoriatic arthritis | Rheumatoid arthritis | Reactive arthritis |
| Osteoarthritis | Reactive arthritis, eg *Yersinia, Salmonella, Campylobacter* | Osteoarthritis | Psoriatic arthritis |
| Trauma (haemarthrosis) | Ankylosing spondylitis | Viruses (eg hepatitis A, B, & C; mumps) | |
| | Osteoarthritis | Systemic conditions* (can be either) | |

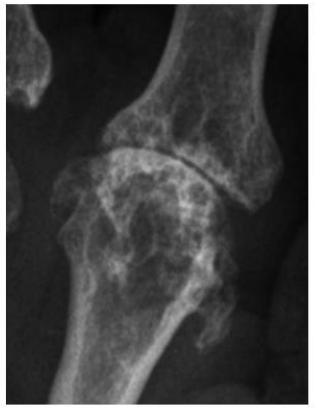
# Differentiation

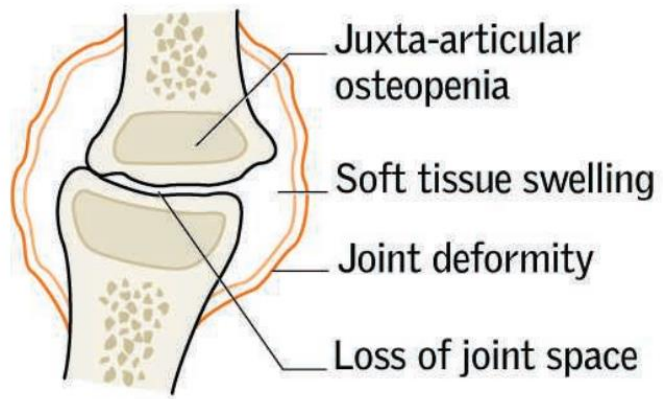
## Osteoarthritis






## Rheumatoid Arthritis



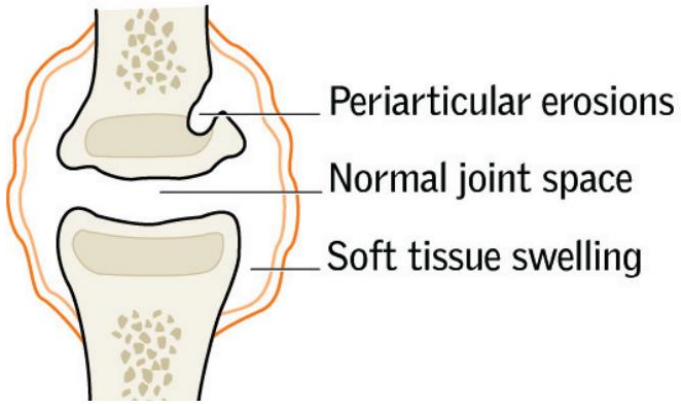





Bamboo Spine
Progressive loss of range of movement.

## Ankylosing Spondylitis






## Gout

# Management

- If you suspect an undiagnosed inflammatory condition, a referral request can be made for rheumatology. This can involve blood tests.
- If a patient has been referred after having had recent scans and blood tests with nothing abnormal detected, it is likely that an inflammatory disease has been considered prior to referral.
- Acute inflammatory autoimmune disorders (like a flare up of rheumatoid arthritis), should be considered with PEACE (protect, elevate, avoid anti-inflammatories, compression with support and education). Although as this is an autoimmune disorder, anti-inflammatories may be prescribed as appropriate.
- Recovery from flare-ups can be managed with low-impact exercise. Consider LOVE, **L**oad appropriately, **O**ptimism, pain free **V**ascular exercise, **E**xercise for mobility, strength and function (as appropriate).
- Progressive disorders such as ankylosing spondylitis should focus on exercise and maintaining mobility and range of movement.

Question: Would joint mobilisations be appropriate for rheumatoid arthritis?

# Rheumatoid Arthritis Summary

Auto-immune disorder
Many joints (usually) – hands, wrists and knees common (pain in big toe. Tenderness, swelling. Bilateral (common)
Fatigue
Fever (low-grade)
Weight Loss
Chronic Pain

2-3 times higher in women
Highest onset in 60s

HLA – human leukocyte antigens

Increased risks from: Obesity, smoking, Mothers Smoking, lower income parents

Palpation: Joints warm to touch

Management: Manage pain during flare-ups, mobility aids, adaptations. LOVE after a flare-up has settled.

# Axial Spondyloarthritis Summary

Axial Spondyloarthritis (axSpA)
Inflammatory Arthritis – non-radiographic axSpA and Ankylosing Spondylitis

**Ankylosing Spondylitis:**
Progressive, Debilitating. Early diagnosis is key.
Affects the spine and sacroiliac joints (mainly)
Plantar Fasciitis, Achilles Tendonitis, Costochondritis – resulting in chest pain (Enthesitis)
Inflammation of entire digit (Dactylitis)
Extra-articular: iritis (anterior uveitis) – 1/3 of patients, psoriatic arthritis, inflammatory bowel disease.

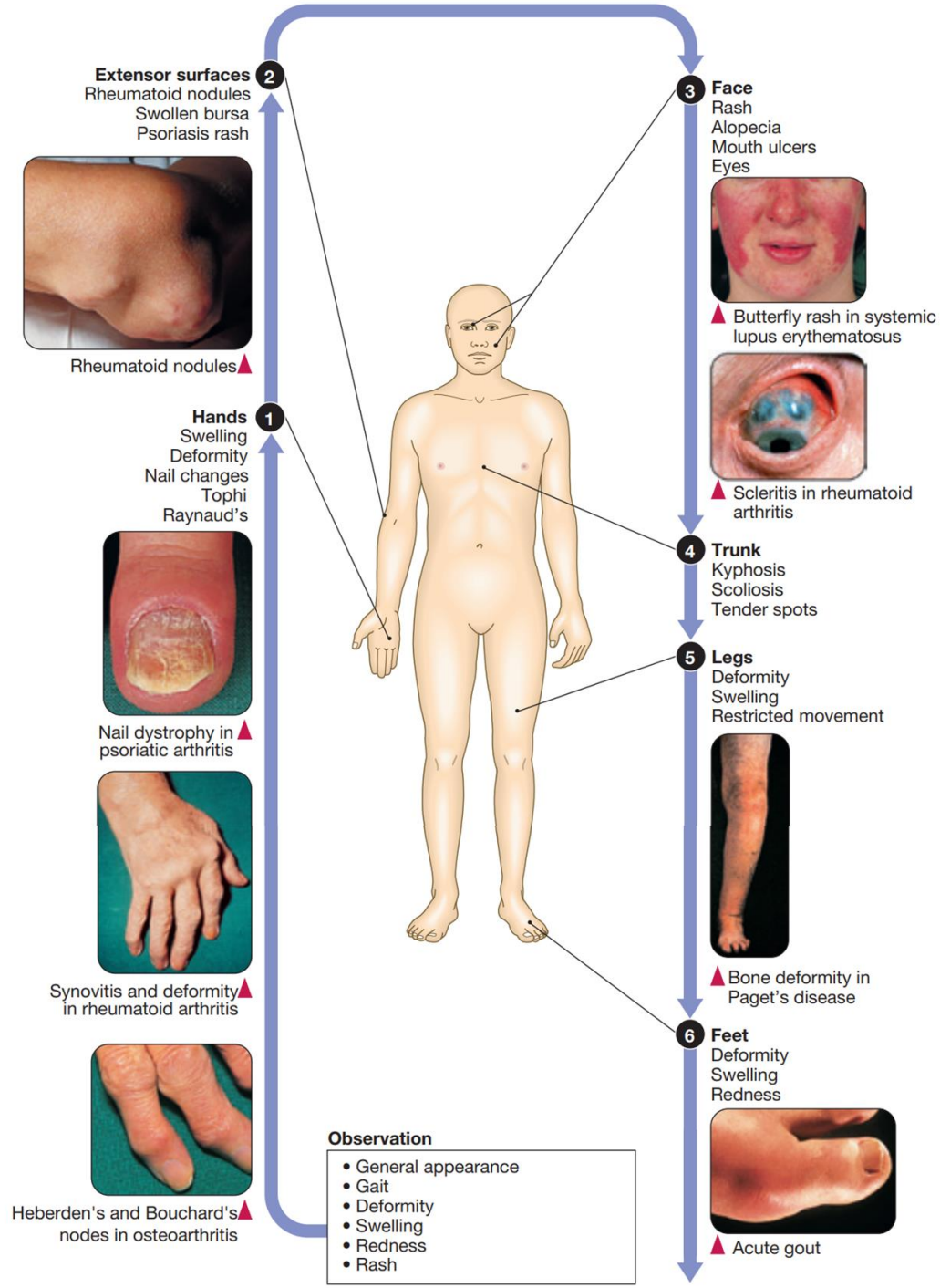
6 times higher in males at 16 years old, and 2 times higher in males at 30 years old.

Typical patient: male < 30 years old, gradual onset of back pain, worse during the night with spinal stiffness. Relief through exercise. Improves by end of day.

Disease course is variable. Later stages calcification of ligaments results in a ‘bamboo spine’ appearance on X-rays

Management: Exercise, not rest for back pain. Exercise programmes to maintain posture and mobility (including thoracic expansion).

Summary of What to Observe:



Rheumatoid arthritis (RA),

Axial Spondyloarthritis

Ankylosing spondylitis (AS)

Psoriatic arthritis (PsA),

Fibromyalgia

Polymyalgia

Systemic lupus erythematosus (SLE, lupus)

Sjogrens

Gout



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